§1.56(11): projective ⟹ entire relation contains a map

R is an entire relation from A, tabulated as ⟨T; x: T ⟶ A, y: T ⟶ B⟩.
A is projective: the cover x has a left inverse z : A ⟶ T with z · x = 1_A.
Then z is the §1.413 containment witness: **graph(z · y)** ⊂ **R**.

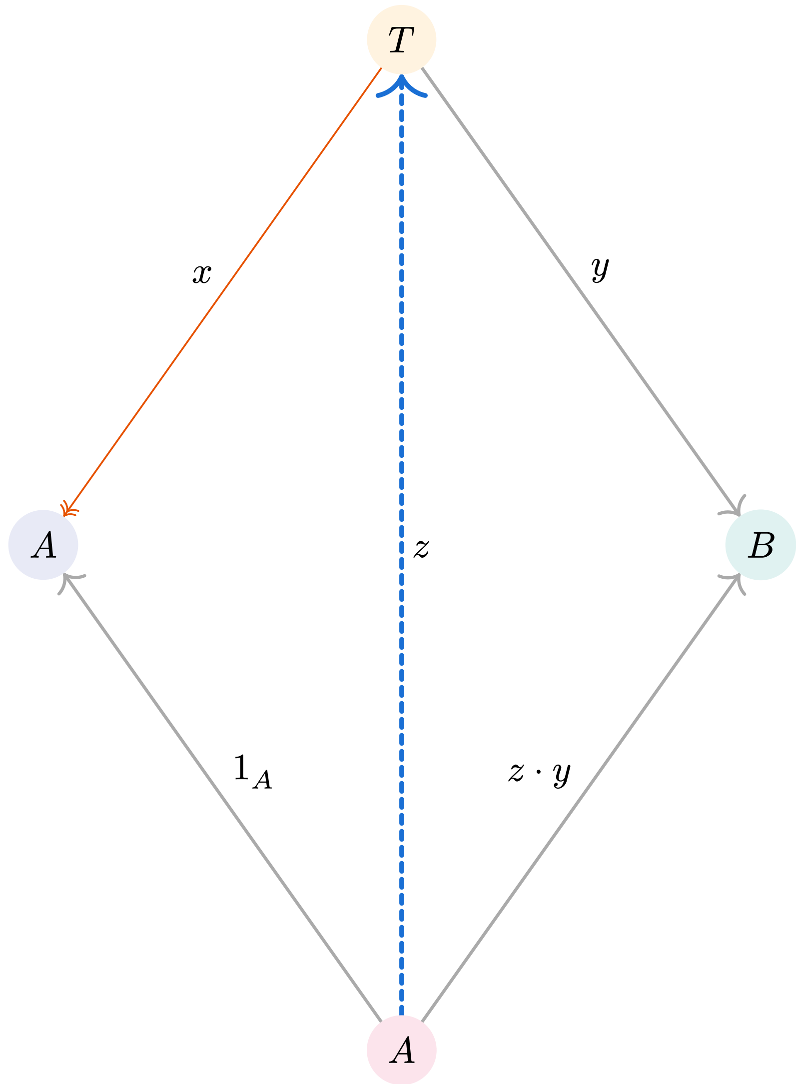

Figure 1: **§1.413 containment:** the dashed arrow z is the (unique, monic) witness ⟶ $z \cdot x = 1_A$ and $z \cdot y = z \cdot y$
⟶ $z \cdot x = 1_A$ makes z a **split monic**
⟶ so z is monic, matching §1.413's "necessarily monic"
Left: columns of the tabulated relation R.
Right: columns of the contained graph graph$(z \cdot y)$.

§1.413: containment witness

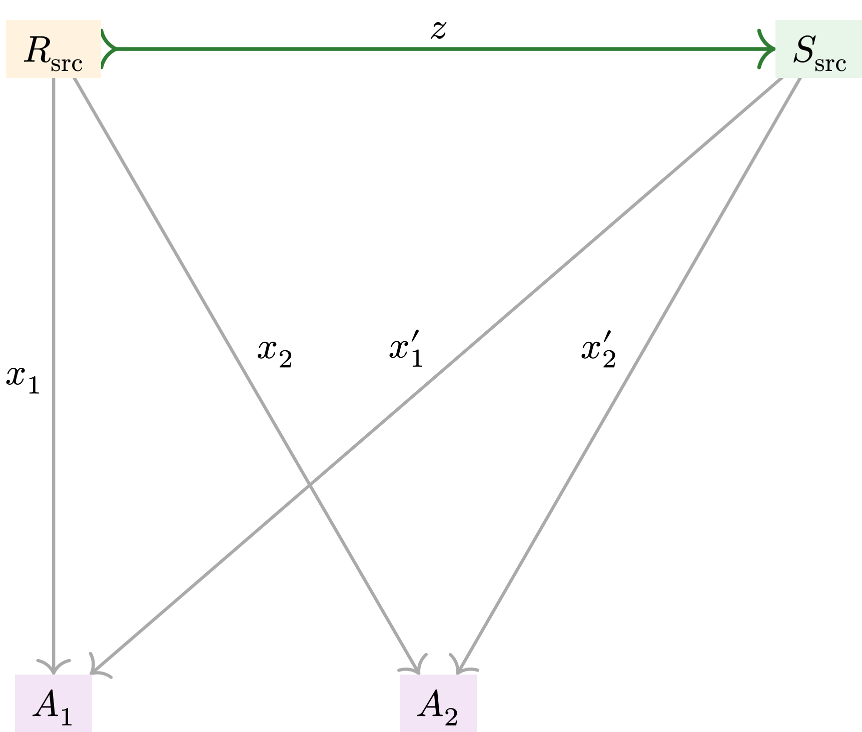

Figure 2: General definition: $(T; x_i) \subset (T'; x'_i)$
$\Leftrightarrow \exists z : T \to T'$ with $z \cdot x'_i = x_i$ for all i.
z is **unique** (because the x'_i are a monic family) and **necessarily monic** (because $z = z'$ when post-composed with a monic family).

Formalized in `Fredy/S1_56.lean`

`projective_entire_contains_map` (⟹)
`entire_contains_map_projective` (⟸)
`RelHom_unique` — witness is unique
`RelHom_monic` — witness is monic